Network Operations Strategy Memo

To: Head of Network Operations, Delhivery **From:** Data Science team **Re:** Graph-based ETA intelligence — findings and recommended interventions **Date:** June 2026 | Length: 2 pages + technical appendix

The problem, quantified

OSRM, our current ETA engine, is wrong almost everywhere. Across ~26,300 corridor segments analyzed, the **median actual transit time is 1.9× the OSRM estimate**, and **91% of segments exceed OSRM by more than 20%** — our SLA-breach threshold. OSRM's mean error is **92 minutes per segment**, and only **7% of trips** land within 15% of the promised time. This is not a tuning problem; the engine structurally ignores dwell time, congestion, and network position.

What we built

We modeled the network as a directed graph — 1,657 facilities as nodes, 2,777 corridors as edges — with edge weights set to each corridor's historical actual-vs-OSRM delay ratio. This structure feeds two things: (1) a graph-enhanced ETA model, and (2) a structural audit of where delay actually originates.

On a held-out test set of 7,388 segments:

Model	MAE (min)	Within 15% of actual
OSRM (current)	91.8	7%
Baseline regression (trip features only)	42.8	36%
Graph-enhanced (corridor history + centrality + node2vec)	**32.8**	**50%**

The graph model cuts error by **23% vs. an already-strong baseline** and **64% vs. OSRM**, and triples the share of trips OSRM gets "right." The single most valuable graph feature is the corridor's historical delay ratio — proof that delay is a *stable property of network position*, not random noise. It can be predicted, and therefore promised against.

Top 5 bottleneck hubs (by SLA-breach contribution)

Hub	% of all late segments originating here	Median delay ratio	Why it matters
Gurgaon_Bilaspur_HB (Haryana)	3.9%	1.69×	Highest betweenness in the network (0.24) — delays here cascade across the network
Bhiwandi_Mankoli_HB (Maharashtra)	3.4%	2.16×	Mumbai's gateway; its corridors to Mumbai Hub and Mira Road run late on **100% of trips**
Bangalore_Nelmngla_H (Karnataka)	2.9%	1.61×	Anchors the chronically late Bengaluru intra-city triangle (KG Airport, Bommasandra)
Pune_Tathawde_H (Maharashtra)	1.6%	1.75×	Pune→Bhiwandi corridor late on 100% of 107 trips — compounds the Bhiwandi problem
Bengaluru_Bomsndra_HB (Karnataka)	1.5%	1.50×	Second leg of the Bengaluru triangle; bidirectional chronic delay

Together the top 3 hubs originate **~10% of all late segments network-wide**; the top 5, **~13%**. Note the geography: this is not 1,657 small problems — it is **three metro clusters (NCR, Mumbai, Bengaluru)** plus the corridors between them.

Recommended interventions

1. Bhiwandi–Mumbai cluster: add a parallel route + dwell audit (highest ROI). Three corridors out of Bhiwandi_Mankoli run late on literally every observed trip, with median delays of 2.2–2.8×. Delay this consistent is *not traffic* — traffic varies. It is process: dock congestion, batching policy, or cutoff scheduling. Action: a 2-week time-and-motion audit at Bhiwandi, plus piloting a direct Mumbai_Chndivli→Mumbai Hub route bypassing Bhiwandi for time-sensitive freight.

2. Bengaluru triangle: re-time, don't re-route. The Nelamangala–Bommasandra–KG Airport triangle is chronically 1.4–1.7× late in both directions. Our time-of-day cells show evening departures from high-centrality hubs carry the worst penalty. Action: shift cutoffs on these three corridors out of the evening window and re-promise SLAs using the graph model's corridor-specific ETA rather than OSRM.

3. Gurgaon_Bilaspur: protect the network's keystone. It is only ~1.7× late — but it touches more shortest paths than any other facility. A 10% capacity disruption here propagates further than a 50% disruption almost anywhere else. Action: prioritize it for capacity headroom investment and make it the first hub on real-time delay-risk monitoring.

4. Route-type policy: distance-banded FTL/Carting rules. The data overturns the intuition that FTL is always faster: **Carting outperforms FTL beyond 300 km** (0.97× vs 1.72× median ratio), while **FTL wins under 100 km** (1.81–2.15× vs 1.92–2.32×). The largest single saving: 100–300 km afternoon departures from mid-centrality hubs, where switching Carting→FTL cuts the delay ratio from ~7.0× to ~2.0×. We deliver this as a lookup framework (distance band × time of day × hub centrality tercile).

Projected impact

If the top 3 hubs are brought from their current delay ratios to the network median (1.9×), we estimate a **~10% reduction in late segments network-wide** — and because these hubs sit upstream of high-volume corridors, the customer-facing improvement is larger than the raw count suggests. Replacing OSRM promises with graph-model ETAs would, **with zero physical investment**, move the share of accurate promises from 7% to 50%, directly reducing SLA penalty exposure on the ~91% of shipments that currently get an unrealistic promise.

The sequencing we recommend: deploy the graph-model ETA for promising (weeks, software-only) → Bhiwandi dwell audit (month 1) → Bengaluru cutoff re-timing (month 2) → Bilaspur capacity review (quarter).

Methodology & assumptions (technical appendix)

For the analytics team — not required reading for the operational recommendations above.

Unit of analysis. The raw data is one row per scan checkpoint (144,867 rows). The meaningful unit is a *segment* — a single hop between two facilities — so rows are aggregated by (trip, source, destination), summing the per-scan increment columns. This yields 26,368 segments.

Cleaning. Segments with zero or negative durations, hops under 0.5 km (scan artifacts), and delay ratios outside the 0.2–20 range (timestamp errors) were removed — 100 rows, roughly 0.4% of the data, too few to bias the results.

Core metric. Delay ratio = actual time ÷ OSRM time. A segment is flagged "late" when this exceeds 1.2 (the 20% SLA-breach threshold).

Edge weights. Each corridor's weight is the *median* delay ratio, not the mean. Delay distributions are heavily right-skewed, and the mean would be dragged upward by a handful of extreme trips.

Bottleneck definition. Hubs are scored on three factors — betweenness centrality (computed on delay-weighted paths, so a hub on many slow routes scores higher), median outbound delay, and trip volume. The memo ranks hubs by SLA-breach contribution (share of all late segments originating at the hub) because that reflects volume-weighted operational impact.

Leakage control. The dataset's own training/test split is honored. Every graph statistic — corridor history, centrality, PageRank, and the node2vec embeddings — is computed on training segments

only and then mapped onto test rows. Corridors seen only in the test set fall back to the global training median, so no test information enters the features.

Model. A gradient-boosting regressor predicts actual segment time, modeled as log(1 + time) to handle the skew. The graph model adds corridor history, hub centrality/degree/PageRank, and 16-dimensional node2vec source/destination embeddings on top of the baseline trip features. The historical corridor delay ratio is the single most informative graph feature, confirming that delay is a stable property of network position rather than random noise.

Scope limit. The data covers September–October 2018 only, so seasonal volume spikes are untested.